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VI SEMESTER

B.TECH. [IT]

MID-SEM EXAMINATION

(March- 2020)

IT 306 ARTIFICIAL INTELLIGENCE & Expert Systems

Time: 1:30 Hours

Max. Marks: 20

Note:	Attempt All Questions.

Q1) Solve the following crypt-arithmetic problem by using Generate and Test method, and applying the following constraints: (5)

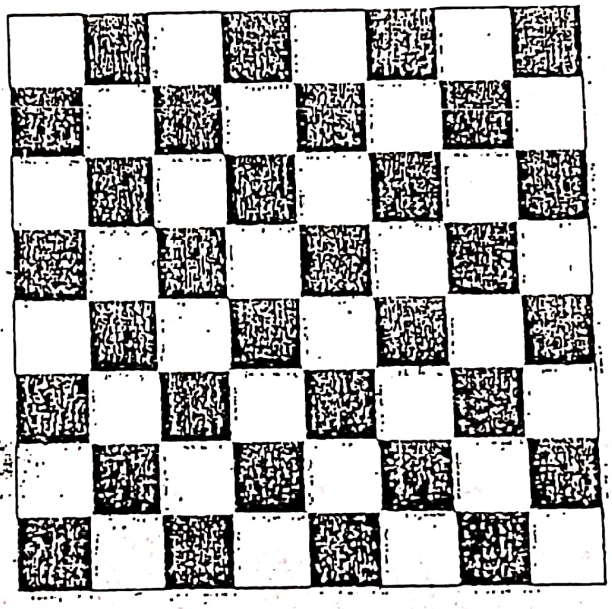
a) Each letter represents exactly one number (in the range [0-9])

b) No two different letters represent the same number

```
    M O O N
+   S O O N
+   N O O N
    -----------
    J U N E
    -----------
```

Q2) Write the steps of A* algorithm? How A* proves to be a better search algorithm then Uniform Cost Search in the same state space? (5)

Q3) Explain the three variants of hill climbing algorithm: simple hill climbing, Steepest-Ascent Hill Climbing and Stochastic hill climbing. Draw state space diagram for hill climbing algorithm with detailed discussion on different regions in the state space diagram. How the problem of local maximum, plateau and ridge can be solved to find out the optimal solution? (6)

Q4) You have a chessboard (8x8) plus a big box of dominoes (each 2x1). Use a marker pen to put an "X" in the squares at coordinates (1, 1)

and (8, 8) - a pair of diagonally opposing corners. Is it possible to cover the remaining 62 squares using the dominoes without any of them sticking out over the edge of the board and without any of them overlapping? You cannot let the dominoes stand on their ends. (4)